

ADDENDUM TI MANUAL CONTAINER SYSTEM DEPLOYMENT



NOTE: All System Manuals Can Be Found in the USER MANUAL Box, Located Directly to the Right of Rack B



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NOTICE

Advanced Navigation and Positioning Corporation has attempted to describe in this manual all areas of possible danger to personnel in connection with the use of this equipment. This document refers to United States government and professional technical standards: Consult government standards prevailing in the region of installation to assure compliance with local requirements.

Personnel should use caution when checking, operating, and servicing this equipment, especially when power is on. Like all electronic equipment, care should be taken to avoid electrical shock in all circuits where substantial currents or voltages may be present, through design or short circuit. Caution should be observed also in lifting and hoisting equipment especially regarding large structures during installation.

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SECTION 1: UNLOADING TRUCK AND CONTAINER FROM PLANE

1. After unchaining the truck, back out from underneath the wing to allow for extra ceiling clearance.
2. Inflate the truck's airbags. This process will take 15 minutes.
3. Proceed to back the truck down the ramp and off the plane.
4. Attach the rear bumper with 8 bolts provided.

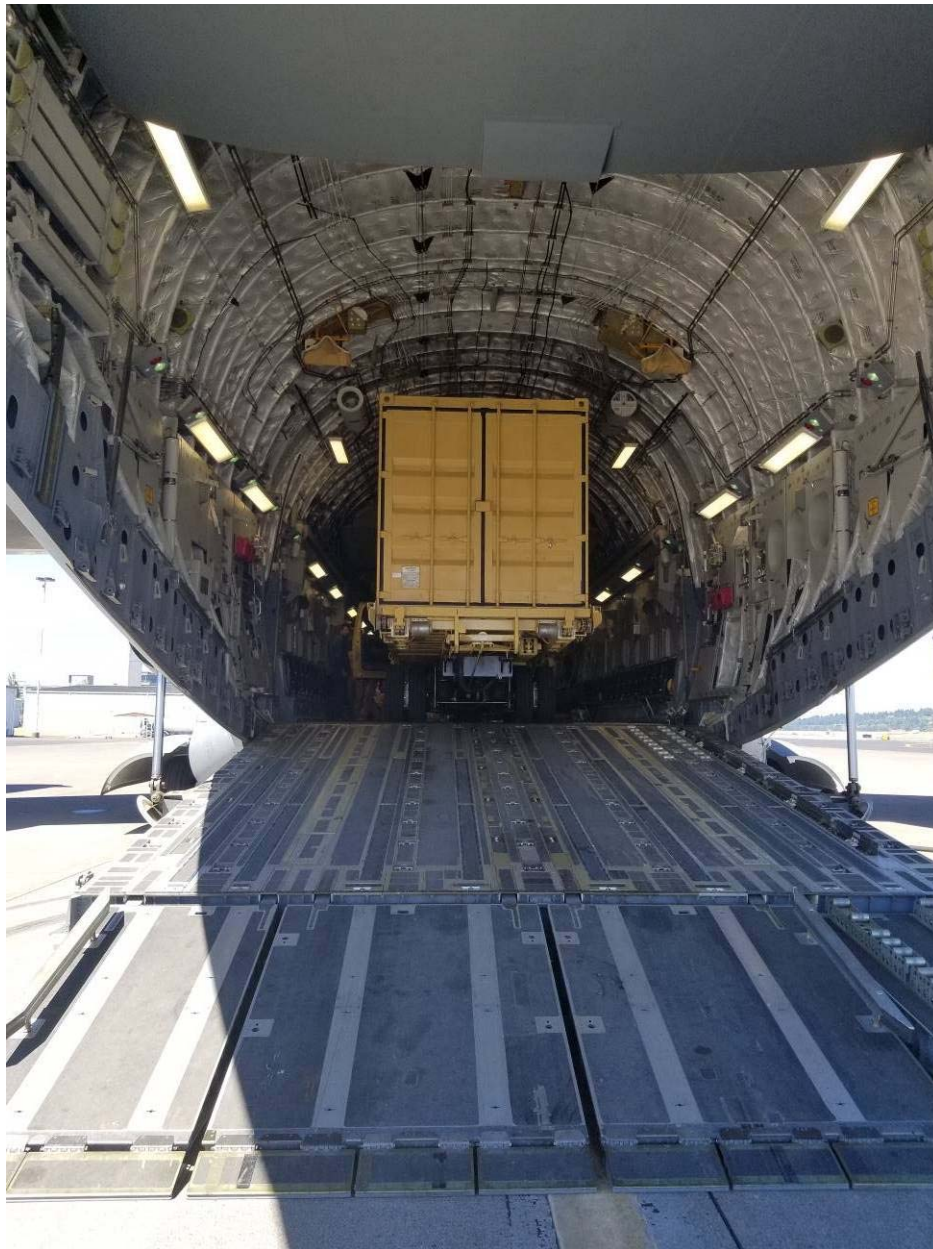


Figure 1 Unloading Truck and Container From the Plane

SECTION 2: UNPACK THE CONTAINER

Carry items directly to where they will be assembled, this saves time. Tripods can be assembled and then carried to the location where intended to be installed.



Figure 2: Partialy Packed TTLS

SECTION 3: UTILITY POWER

3.1 Single Phase 240 vac

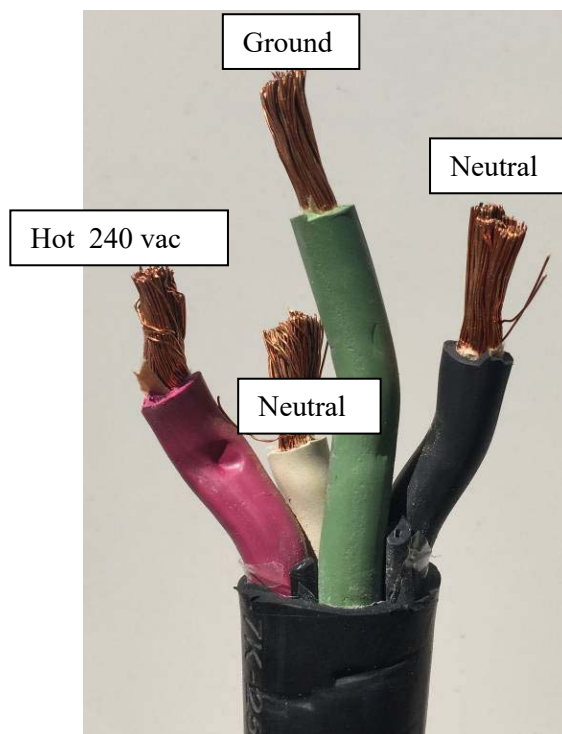


Figure 3 Single Phase 240 vac

3.2 Split Phase 240 vac

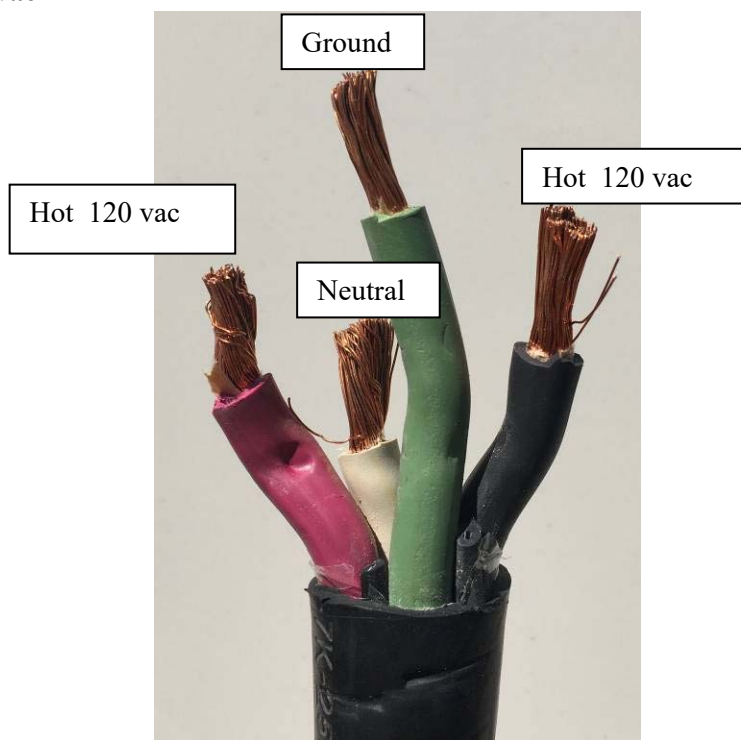


Figure 4 Split Phase Power Cable Wiring

SECTION 4: GENERATOR ASSEMBLY AND EXCERSIZE

4.1 Generator Assembly

1. Use 5 bolts (included with generator) to attach handle to generator.

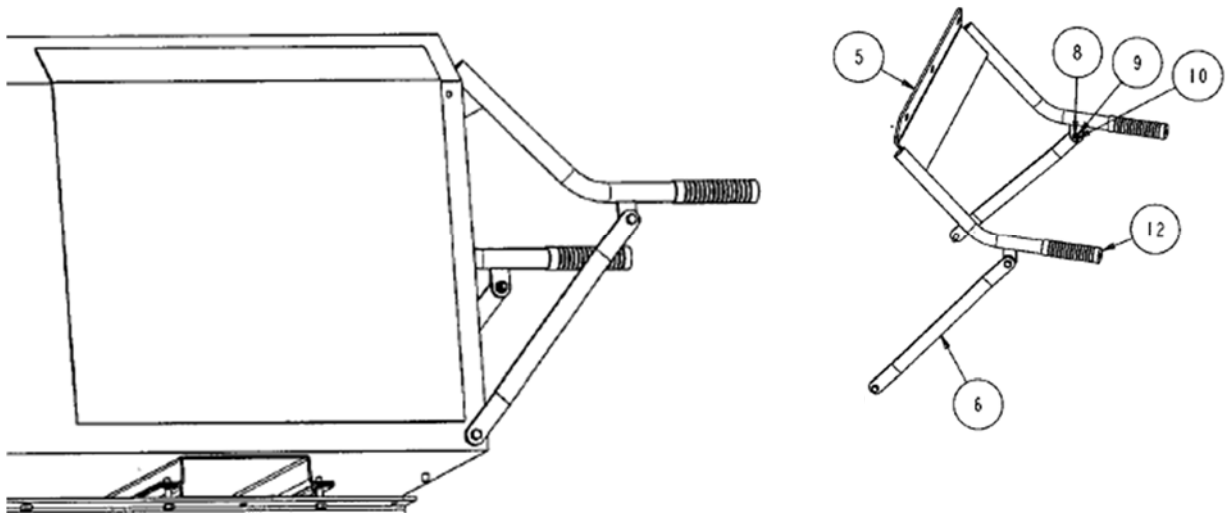


Figure 5 Generator Handle



Figure 6 Assembled Generator

2. Use the yellow ramps to assist with rolling the generator out of the shelter. After unloading the generator from the shelter, attach the switchover and power cables to both the generator and the container.

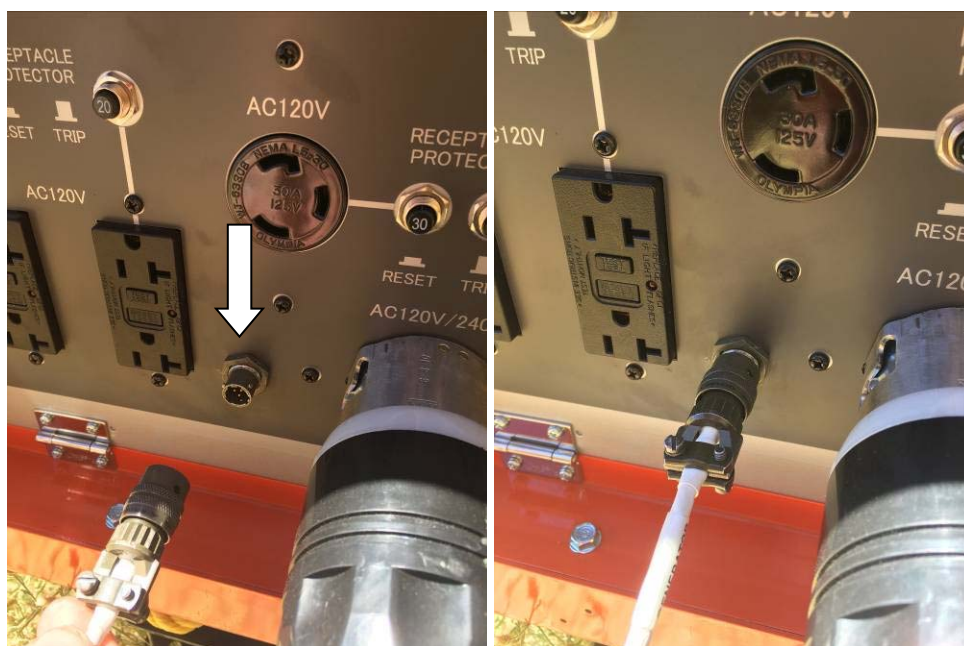


Figure 7 Plugging the Switchover Cable into the Generator

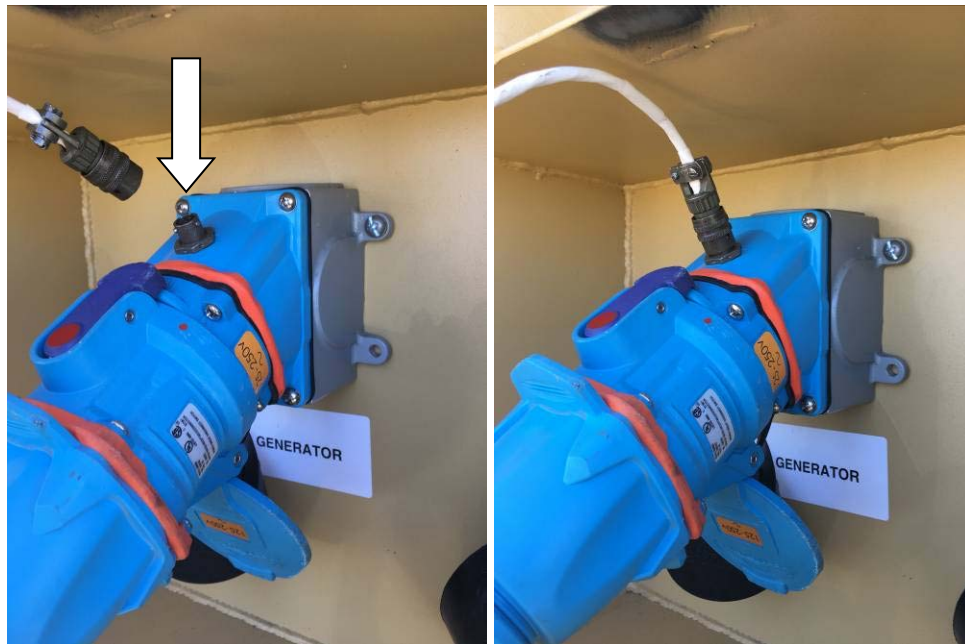


Figure 8 Plugging the Switchover Cable into the Container

3. If the generator is equipped with a fuel drum, attach the drum to the generator in the following manor. Inside of the engine access panel, switch the fuel valve to the drum. The fuel lines are keyed and cannot be swapped.



Figure 9: Generator/ Drum Hook up

4.2 Setting Generator Exerciser

1. To Set Exerciser:

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Press and hold (5 seconds) the Set Engine Exerciser button. The exercise period occurs immediately and at approximately the same time weekly thereafter. The status light below the button blinks during the exercise period. The light stays on to indicate that the exerciser has been set. If the light is off, the exerciser has not been set.

2. To Cancel an Active Exercise Period:

Press the Bypass Time Delay button to stop an exercising generator. If exercise with load is set, the ATS retransfers the load to the utility, then stops the generator after cooldown

SECTION 5: SENSOR INSTALLATION

5.1 ESA Tower

1. Position the base hinge such that a flat side of the tower will face the approach direction. Secure the base hinge with steel stakes provided in the kit. Measure 25 feet from the base plate to the location where each auger will be placed. Twist augers are the preferred anchor for the ESA. Arrow head anchors can also be used in place of twist anchors, but twist anchors are preferred for ease of installation.
2. Note that if arrow head anchors are used, take care to position the entry point of the anchors such that the eyelet will be 25 feet from the base hinge. If using arrow heads, pound arrow head anchors at an angle into the ground at the dimensions shown below. Leave 3 to 5" of anchor cable above ground, and attach chain and turnbuckle.
3. To help with the placement of the anchors, use the guy wire spacing tool, as depicted in. Simply slip the closely attached loops over two base plate attachment points and pull the free end taut.

Maintain tension in both angled sections of rope to ensure alignment, and then mark the location of the loop at the end as the anchor point.



Figure 10 Arrow Head Ground Anchors

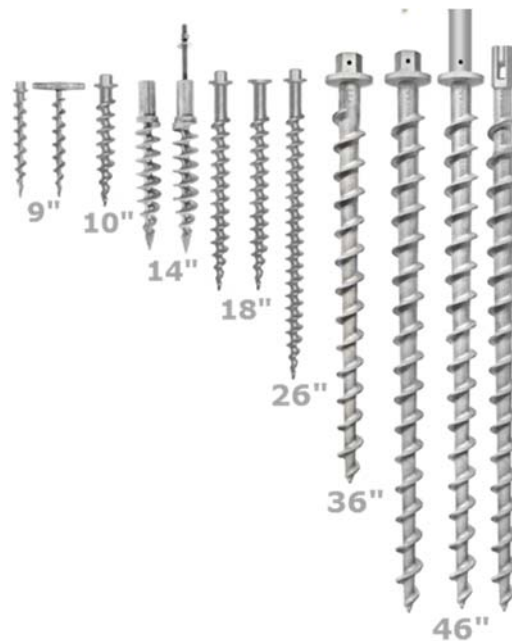


Figure 11 Twist Anchors



Figure 12 Arrow Head Ground Anchor, Chain, and Turnbuckle



Figure 13: American Earth Anchor

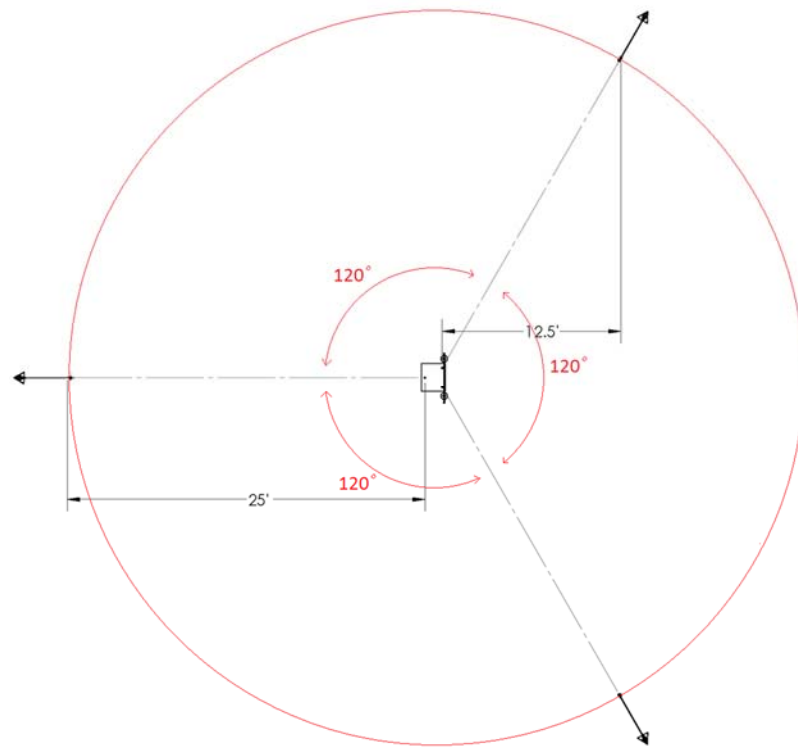


Figure 14 Ground Anchor Placement



Figure 15 Guy Wire Spacing Tool

4. Assemble 3 tower sections and top plate to base plate hinge.



Figure 16 ESA Tower/Hinge Close-up view

5.2 Gin Pole Assembly and Use



Figure 17: Gin Pole Assembly Components

5.2.1 GIN POLE ASSEMBLY

1. Install Upper Leg 1 onto the bracket that is already installed on Upper Leg 2 using two each 1-1/8" Hex X 3/4"-10 Screws and nuts.



Figure 18: Close up of Gin Pole Bracket

2. Assemble Lower Legs into Upper Legs using the locking pins.



Figure 19: Gin Pole Leg Assembly Close Up

3. Screw the claw feet into the bottom of the Lower Legs.



Figure 20: Gin Pole Foot Assembly Close Up



Figure 21: Gin Pole Final Assembly

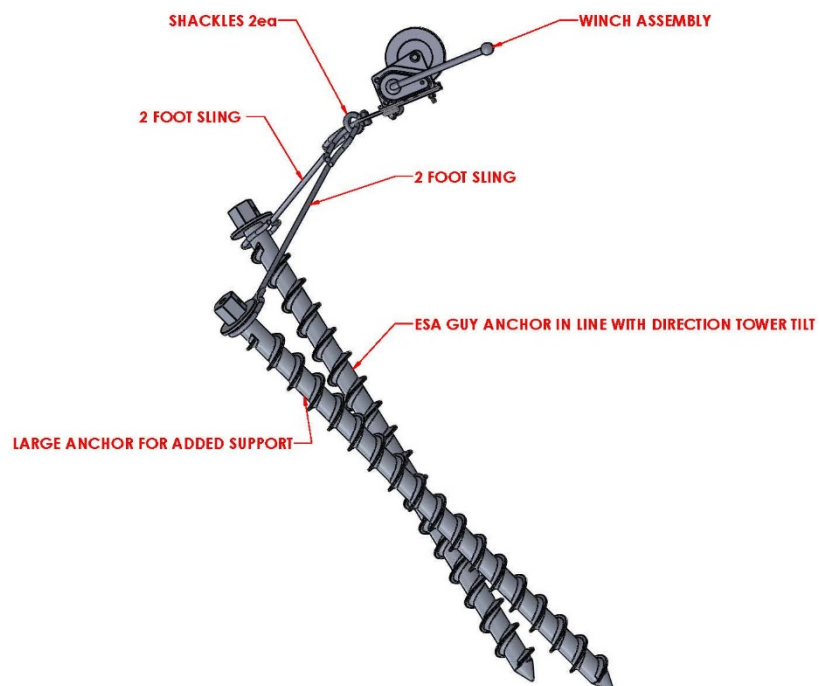


Figure 22: Gin Pole Winch Assembly

5.2.2 RAISING THE TOWER WITH A GIN-POLE

1. Position the tower/hinge so that when the tower is lifted, a flat side of the tower will be facing the threshold.
2. Attach the guy wires to the tower.
3. Anchor the two top guy wires that point toward the threshold. Leave the top and middle guy wires of the guywire that is parallel to the centerline loose.
4. Lay the Gin Pole on the ground with the bracket 4 feet from the anchor that will be used to attach the tower when it is lifted. Use augers to secure the claw feet.



Figure 23: Close up of Anchoring Gin Pole Feet

5. Flip the gin pole over so the bracket points towards the tower.
6. Attach the upper guywire to the winch cable if the tower has been completely assembled with antennas. The middle guywire may be used if the tower has no antennas.
7. Attach the winch to the guy anchor that has no guywire attached. Note, depending on soil conditions it may be necessary to install another anchor to support the raising of the tower.
8. Reel out the winch cable and attach to the bracket side that is facing up.
9. Winch the Gin Pole up so that it is 90 degrees to the ground and the slack is removed from the guy wire.

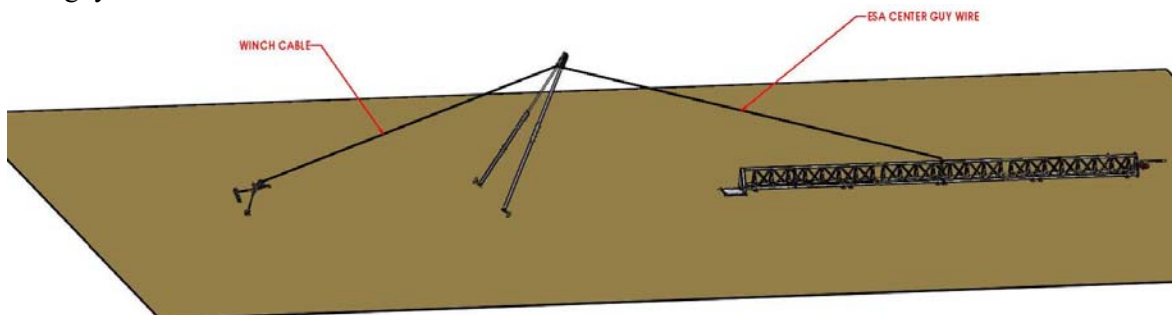


Figure 24: Gin pole lift configuration



Figure 25: Gin pole position at the start of tower lift



Figure 26 - Hand winch assembly

10. Have one person lift the tower while another cranks on the winch.



Figure 27: Lifting the tower



Figure 28: Gin Pole should have all the weight at this step

11. Once the tower is in position. Attach the top guywire to the anchor.

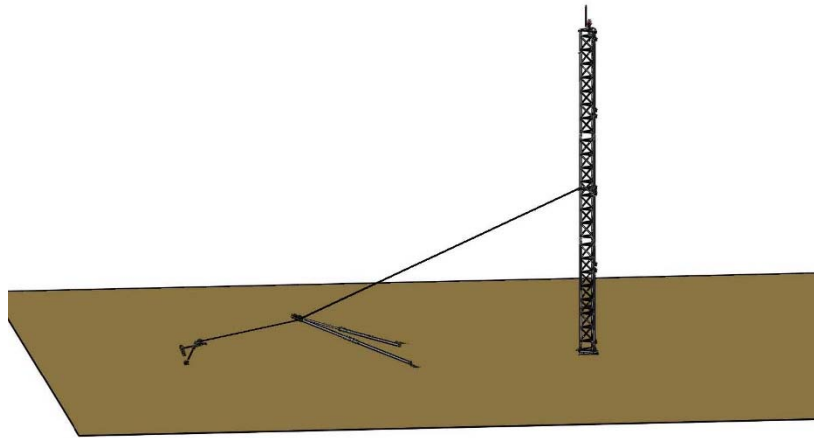


Figure 29: Gin pole position after lifting the tower

12. Disconnect the middle guy wire from the gin-pole and attach to the anchor.

5.2.3 LOWERING THE TOWER WITH A GIN-POLE

1. Remove antennas from the tower and stow in the shelter along with the Sensor and cables.
2. Lay the Gin Pole on the ground with the bracket 4 feet from the anchor that will be used to attach the tower when it is lifted. Use augers to secure the claw feet.



Figure 30: Close up of Anchoring Gin Pole Feet

3. Attach the winch to the guy anchor (guy that is parallel to the runway and opposite the threshold) that will be disconnected to lower the tower. Note, depending on soil conditions it may be necessary to install another anchor to support the raising of the tower.
4. Disconnect the middle guy wire that will be used to lower the tower and attach to the gin-pole bracket.



Figure 31: Gin Pole Placement/ Winch Hook Up for Lowering the Tower

5. Disconnect the upper guywire from the anchor.

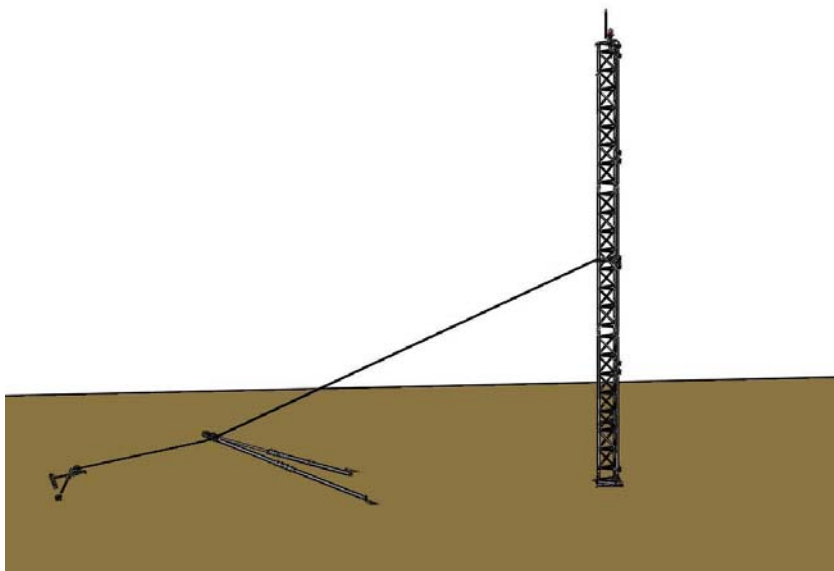


Figure 32: Lowering the Tower – If the tower is preassembled with antennas, use **the upper guywire** to lift

6. Have one person push on the tower, another person lifts the gin-pole at the bracket while another person cranks on the winch counter-clockwise to lower the tower. Once the lines are taut the people on the tower and gin pole should move away from the area while the tower is lowered.



Figure 33: Lowering the Tower

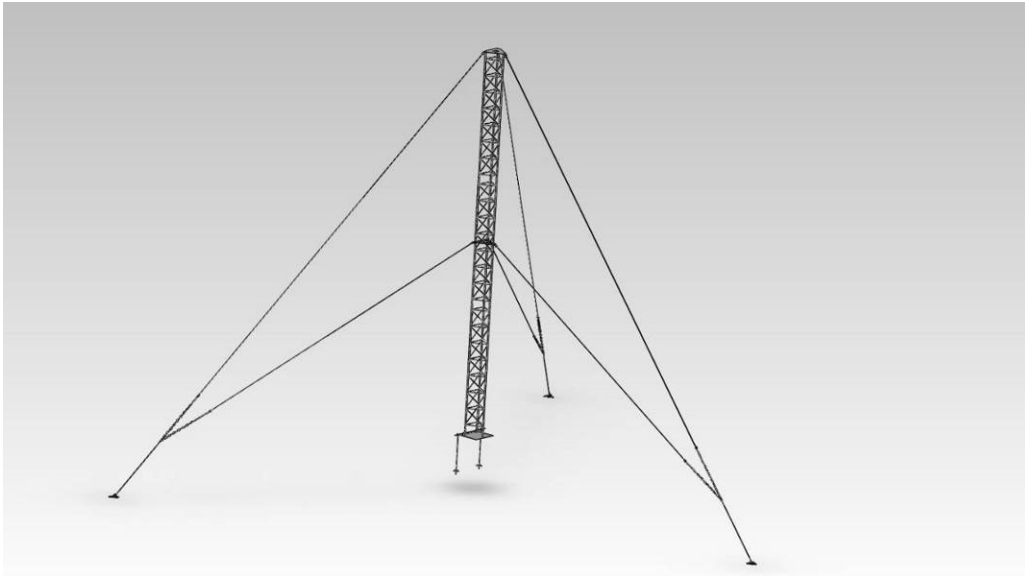


Figure 34: ESA Tower Guywire Configuration

1. Add the guy wire guards to the guy wires.

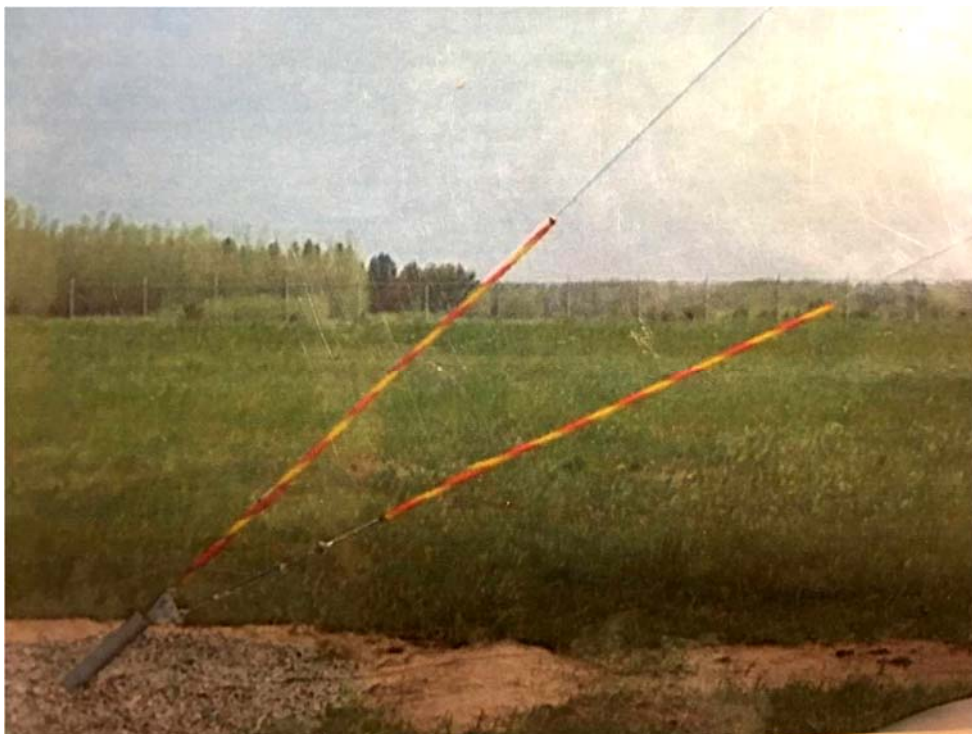


Figure 35 Installed Guy Wire Guards

5.3 Tilt Sensor Installation

1. Using included hardware, install the tilt sensor plate on the inside of the ESA tower, then connect the appropriate cables.



Figure 36 Installed Tilt Sensor Plate

5.4 Antenna Attachment

1. Add antennas starting at the top. Use a rope to support the weight of the antenna while the climber fastens it to the tower with the included hardware.



Figure 37 Attaching Antennas

2. Connect cables to the antennas. Each cable is labeled according to where it should be attached. The cables must only be attached to the correct antenna.
3. Connect the antenna cables to the appropriate port underneath the sensor cabinet, then proceed to connect the cables that run between the sensor and the shelter. Always remember to never pull on the fiber of fiber optic cables, but rather only by the conduit where the metal strain relief is located.



Figure 38 Cable Connection Underneath Sensor Cabinet



Figure 39 Completed ESA Tower

SECTION 6: UPLINK TOWER ASSEMBLY

6.1 Uplink Tower

1. Assemble tower sections on flat ground and attach to hinge at midpoint of shelter.



Figure 40 Uplink Tower on the Hinge



Figure 41 Assemble the Tower Attached to the Hinge

6.2 Hoisting Tower with Winch and Gin Pole

1. Attach the gin pole and L-bracket to the weld plate nearest to the tower. The gin pole should be free to lean away from the tower, but not towards it.
2. Mount the winch and L-bracket in line behind the gin pole. The winch and gin pole can be rotated on their respective L-brackets as needed.
3. Attach the winch hook to a point on the tower that will be approximately the same height of the gin pole when the tower is raised. Keep tension on the winch cable when spooling in any slack to ensure it wraps properly.
4. Attach support ropes to the tower as needed.
5. Raise the tower using the winch.
6. Attach spars before removing the winch hook.



Figure 42 Spars Connected to Tower Hitch Plate



Figure 43 Hitch Installed on Container Corner with Connecting Spar





Figure 44 Spars Supporting Uplink Tower Before Detaching Winch Hook

6.3 Antenna Attachment

1. Attach Localizer and Glide Slope Antennas to the Uplink Tower with the included hardware. The Localizer Antenna is installed above the Glide Slope Antenna.
2. Connect Antenna Cables

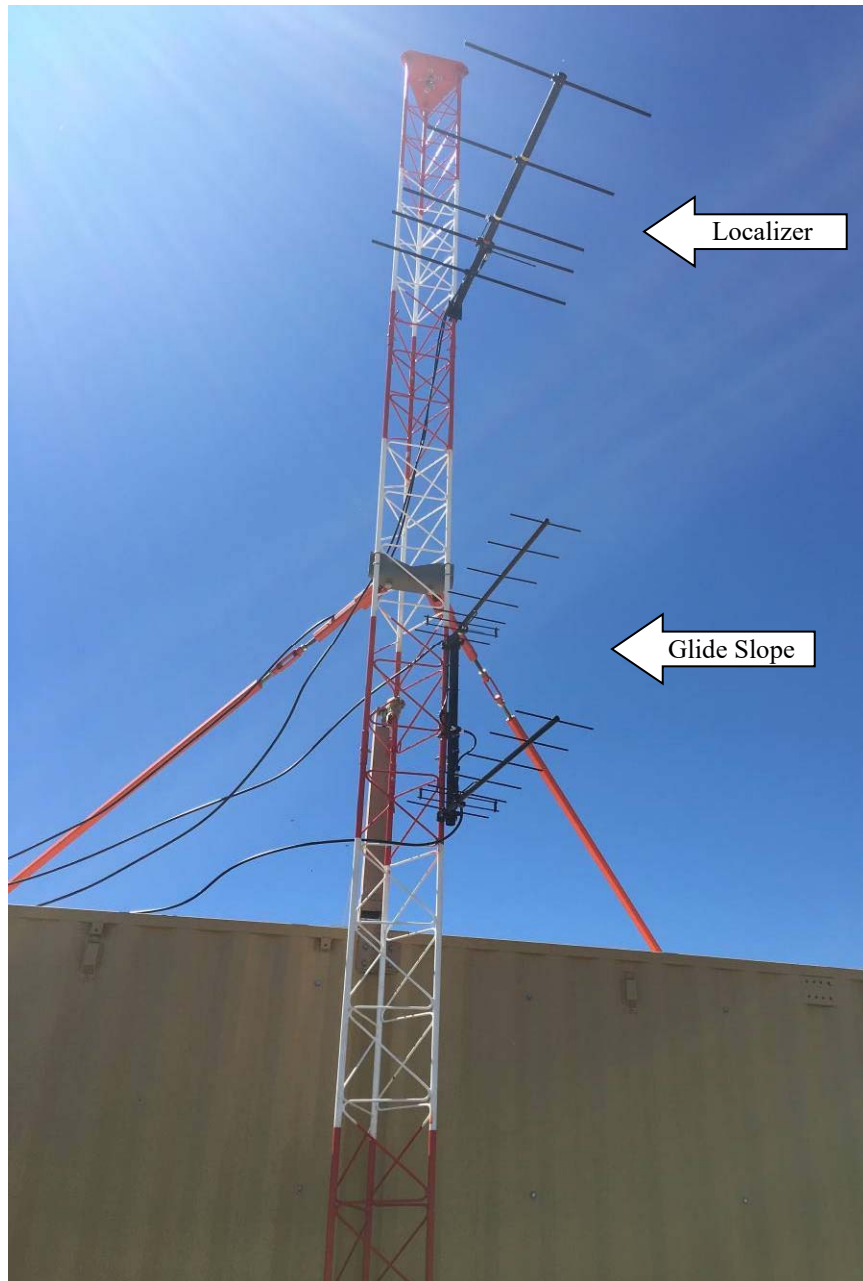


Figure 45 Completed Uplink Tower

SECTION 7: TRIPOD/QUADRAPOD ASSEMBLIES

1. Use included hardware to assemble each tripod (for ATA, Cal/BIT, and Interrogator) and quadrapod (for ASA).



Figure 46 Tripod Assembly



Figure 47 Quadrapod Assembly

2. Fasten brackets for the Cal/BIT and LNA to the Cal/BIT and ATA tripods, respectively.



Figure 48 Bracket for Cal/BIT or LNA

3. Install the Cal/Bit and LNA, making sure that space directly beneath where the antenna will be mounted is unobstructed.



Figure 49 Installed Cal/BIT

4. Install 1 twist anchor each at the installation location for each tripod. Installation of a twist auger normally requires using a steel bar through the auger eye to twist the auger while another person hits the top of the auger with a sledge hammer. Move each tripod over the twist anchor and adjust level using the crank handles on the adjustable legs. Bubble level measurement of the degree of level is all that is required. Secure with a ratchet strap.



Figure 50 Twist Ground Anchor



Figure 51 Tripod with Ratchet Strap

5. Use included hardware to attach antenna to each tripod.
6. Connect all appropriate cables to the antennas, taking care to hang the ends by the strain relievers so as not to damage the cables.
7. Connect the other end of the antenna cables to the corresponding ports on the shelter.



Figure 52 Hang Cables from Strain Reliever Hook and Connect Copper Ground Wire



Figure 53 Strain Relief Hook to Be Installed on Rain Hood to Suspend Fiber / Power Cables

SECTION 8: FIBER/POWER CABLES

- Never pull the cable by gripping the fiber optic connector.
- Always pull the cable toward the assembly where the cable will be attached by gripping the strain relief and gently supporting the fiber optic connector to avoid damage.
- When unrolling the cable from the reel, slowly detach the cable end from the reel so as to avoid damage to the cable end.



Figure 54

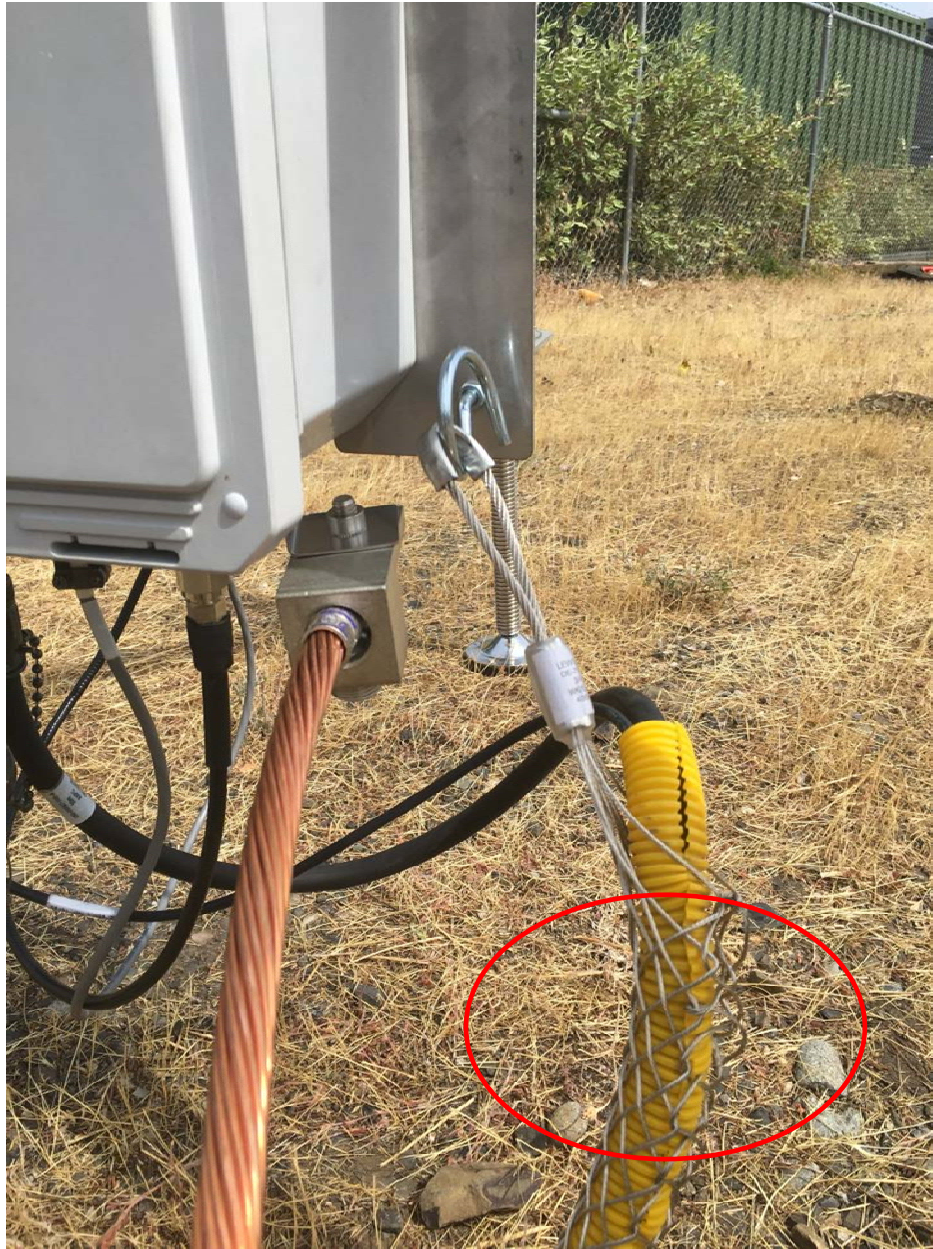


Figure 55 When pulling the cable toward the assembly (ATA, ASA or Cal/BIT), grip the cable by the strain relief (red circle) and be sure that you **DO NOT PULL ON THE FIBER OPTIC CONNECTOR**

SECTION 9: GROUNDING

1. Use hammer to install grounding rods next to the ASA, ATA, Cal/BIT, Shelter and ESA
2. Attach ground cables as shown



Figure 56



Figure 57

SECTION 10: AIR CONDITIONING POWER CONNECTION

10.1 Option 1: Window Units



Figure 58 Attach the Air Conditioners – Outside View



Figure 59 Attach the Air Conditioners- Inside View

10.2 Option 2: Split Units

If the system is configured to have a split unit air conditioner and a truck, the AC units will be installed on the container. If no truck is included or the container is shipped general freight, then the A/C units will need to be hooked up. The attachments are all keyed and the refrigerant lines only reach to their respective attachment points.

10.3 Option 3: No Air Conditioning

For those customers operating in colder climates, such as Antarctica, space heaters have already been installed in the container to maintain a suitable ambient temperature for the electronics.

Appendix A Acronyms & Abbreviations

Acronym	Definition
ANPC	Advanced Navigation and Positioning Corporation
ASA	Azimuth Sensor Array
ATA	Alternate Time-of-Arrival Array
Cal/BIT	Calibration/Built-in-Test
ESA	Elevation Sensor Array